

CYBERSECURITY

VIRTUAL ENVIRONMENT CONFIGURATION

1. Default configuration

To perform the tasks presented in the LAB7 - LAB14 lists it is necessary to use several different machines. The default configuration of the required environment requires at least 3 virtual machines:

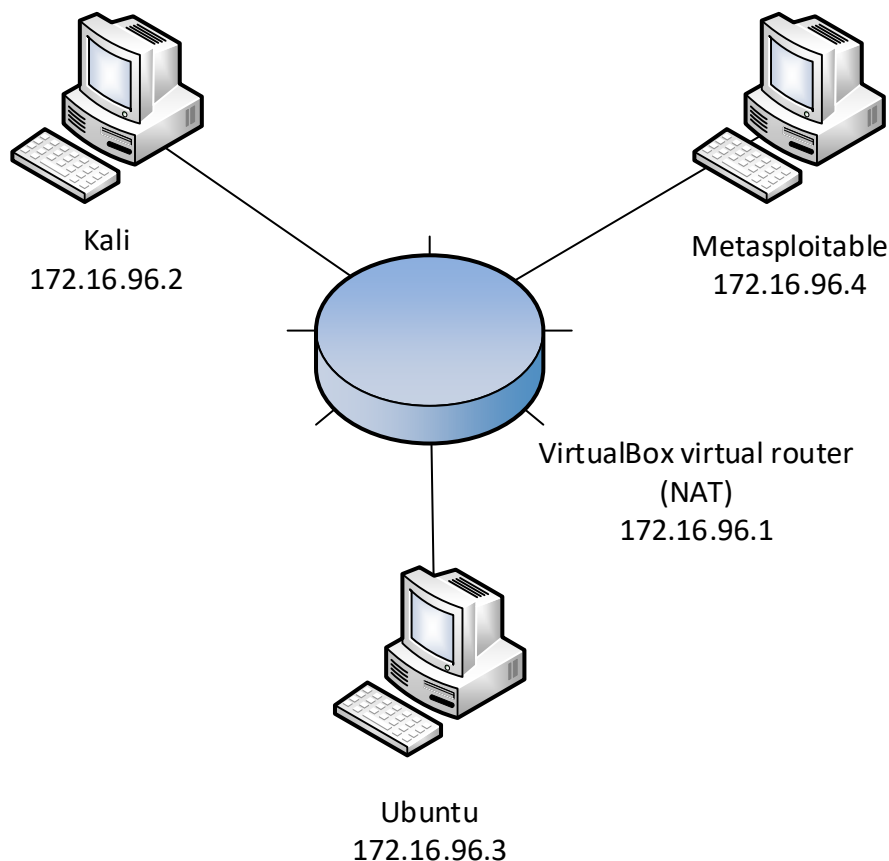
- **Kali Linux** - A virtual machine simulating a computer used by a hacker to perform an attack and VPN server.
Metasploitable 2 - A virtual machine with multiple security vulnerabilities and a range of services in versions for easy exploitation and vulnerability demonstration. In laboratory scenarios it will usually be treated as a victim-server.
- **Ubuntu** - A machine simulating a VPN client, the VM contain OpenVPN, Wireshark and OpenSSL. It is nice to import two Ubuntu VMs, because it provides us opportunity to simulate typical VPN network. The VM contains configuration OpenVPN files for both clients A and B.

In order to ensure proper communication between all machines, it is necessary to create a virtual NAT network in VirtualBox (this process is described in the instructions for LAB1).

The correct configuration can be checked by starting all VMs and checking that their IP addresses belong to the same network and that it is possible to connect in any direction with a **ping** command.

The example configuration shown will ensure the creation of the network shown in the figure below:

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2. Virtual machines used in exercises.

Most laboratories only require the use of a limited number of virtual machines, these requirements are presented in the table below:

LAB NUMBER	REQUIRED VIRTUAL MACHINES
LAB 6	Kali, Ubuntu
LAB 7	Ubuntu client
LAB 8	Kali, Metasploitable 2, Ubuntu
LAB 9	Kali, Ubuntu A, Ubuntu B (optionally)
LAB 10	
LAB 11	Kali, Metasploitable 2
LAB 12	
LAB 13	Kali
LAB 14	Kali